Roll Number | | | | | | | | | | | | |

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
EE(AM)201 - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

TIME ALLOTTED: 3Hours **Maximum Marks : 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.	Power Factor of an AC circuit is	**(1)**	CO1	BL1	PO1
	(a) Tangent of the phase angle (b) Cosine of the phase angle (c) Sine of the phase angle (d) None of the above				
2.	If a circuit does not contain any source of energy or emf, it is known as	**(1)**	CO5	BL1	PO1
	(a) active circuit (b) passive circuit (c) bilateral circuit (d) unilateral circuit				
3.	Three resistance 15 Ω, 25 Ω and 60 Ω are connected in series across 200 V. What will be the voltage drop across 15 Ω ?	**(1)**	CO1	BL2	PO3
	(a) 30 Volt (b) 50 volt (c) 100 volt (d) 180 volt				
4.	The product of V, I and cos θ is called	**(1)**	CO1	BL1	PO1
	(a) active power (b) true power (c) apparent power (d) reactive power				
5.	In semiconductors normally Electrons exist in …….	**(1)**	CO6	BL2	PO1
	(a) Forbidden band (b) Conduction band (c) Valence band (d) Both valence and conduction band				

6.	The Cut in Voltage of a Si diode is	**(1)**	CO3	BL1	PO1

(a) 0.6 V (b) 0.1 V
(c) 6 V (d) 0.7 V

7.	Impure semiconductors are known as …….	**(1)**	CO3	BL1	PO1

(a) compound (b) extrinsic
(c) doped (d) intrinsic

8.	A BJT is a	**(1)**	CO3	BL1	PO1

(a) current-controlled device (b) voltage-controlled device
(c) Impedance controlled device (d) None of the above

9.	What is the number of primary turns in a 400/1000 V transformer if the emf per turn is 10V?	**(1)**	CO2	BL2	PO3

(a) 20 (b) 40
(c) 80 (d) 100

10.	If α=0.98, then the value of β of transistor is:	**(1)**	CO3	BL3	PO2

(a) 49 (b) 0.49
(c) 50 (d) 0.5

11.	Hysteresis loss in a transformer varies as	**(1)**	CO2	BL2	PO1

(a) square of frequency (b) square root of frequency
(c) frequency (d) reciprocal of frequency

12.	Which of the following according to KVL must be zero?	**(1)**	CO1	BL2	PO2

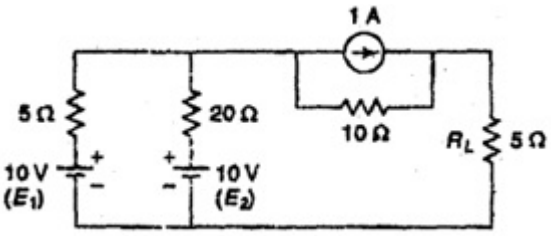
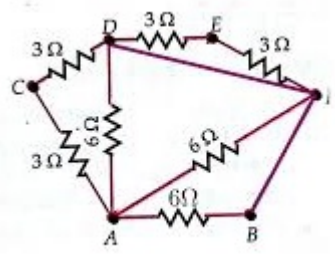
(a) Algebraic sum of voltages in closed-loop (b) Algebraic sum of currents in closed-loop
(c) Algebraic sum of losses in closed-loop (d) Algebraic sum of power in closed-loop

Part B
(Answer any three of the following) 3x5=15

13.	Explain mass action law.? Explain different types of extrinsic semiconductor with examples.	**(5)**	CO3	BL1	PO1
14.	A circuit takes a current of 4 A at 200 V. The current leads the applied voltage by 45°. Find the circuit parameters and draw the impedance triangle.	**(5)**	CO1	BL2	PO3

15.	What is the significance of form factor in ac circuit? What is Peak factor?	**(5)**	CO1	BL1	PO1
16.	Derive the E.M.F. equation of a D.C. Generator.	**(5)**	CO2	BL2	PO1
17.	The emf per turn for a single phase 440/220V, 60Hz transformer is approximately 15V.Find i) the number of primary and secondary turns ii) the net cross sectional area of the core for a maximum flux density of $1Wb/m^2$.	**(5)**	CO2	BL3	PO2

Part C
(Answer any three of the following) 3x15=45

18.	Explain Adder and Differentiator circuits using OPAMP.	**(15)**	CO4	BL2	PO1
19.	Find out the current through the load 5 ohm.  Find out the equivalent resistance of the following figure.  	**(15)**	CO1	BL4	PO3
20.	Define efficiency? What is Auto-transformer? Write the difference between ideal transformer and practical transformer. Derive emf equation of a Single phase transformer.	**(15)**	CO2	BL1	PO1
21.	Write short notes on: (any three)				
	(a) V-I characteristics of Zener Diode	**(5)**	CO3	BL1	PO1
	(b) Inverting Amplifier	**(5)**	CO4	BL1	PO1
	(c) Pin Configuration of OPAMP	**(5)**	CO4	BL1	PO1

(d)	Fixed Bias of BJT	**(5)**	CO3	BL1	PO1
(e)	Voltage Regulation	**(5)**	CO2	BL1	PO1
22. (a)	Explain the concept of Fermi energy level.	**(5)**	CO3	BL2	PO2
(b)	Discuss the effect of temperature on Fermi level in intrinsic semiconductors.	**(5)**	CO3	BL2	PO2
(c)	Compare n-type and p-type semiconductors using energy band diagrams.	**(5)**	CO3	BL4	PO4